

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**Applied Business Analytics**

**M2 Practical 7**

|                             |                                  |
|-----------------------------|----------------------------------|
| Roll No.: T050              | Name: Aashu Yadav                |
| Class: TYIT                 | Batch: 1                         |
| Date of Assignment: 3-10-26 | Date/Time of Submission: 3-10-26 |

**7. Sampling and Sampling Distributions:**

**Part A: Survey and Sampling**

**1. Select a real-life business topic for conducting a survey.**

**Topic:** Social Media Awareness for Bird Rescue and Community-Based Organizations.

**2. Prepare a questionnaire containing at least 5 questions related to the selected topic.**

A questionnaire containing questions related to bird rescue and social media awareness was prepared.

**3. Collect responses from at least 50 people.**

Total 45 responses were collected through the survey.

**4. Store the collected data in a CSV file or Pandas DataFrame.**

**5. Display the first five records of the collected data.**

**Code:**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.read_csv("Bird Rescue Survey .csv")
print("FIRST FIVE RECORDS:")
print(df.head())
```

**Output:**

| FIRST FIVE RECORDS: |                    |                     |          |   |
|---------------------|--------------------|---------------------|----------|---|
|                     | Timestamp          | Name                | Age      | \ |
| 0                   | 9/18/2026 22:05:21 | Radha               | 19 - 25  |   |
| 1                   | 9/18/2026 22:10:56 | Lavkush Vishwakarma | 19 - 25  |   |
| 2                   | 9/18/2026 22:12:12 | Khushi              | 19 - 25  |   |
| 3                   | 9/18/2026 22:12:31 | Priti Yadav         | 19 - 25  |   |
| 4                   | 9/18/2026 22:33:06 | Rani                | 25 Above |   |

**6. Identify the population and sample for the survey.**

**Code:**

```
print("\nTotal Population/Responses:", len(df))
```

**Output:**

Total Population/Responses: 45

**7. Identify the sampling technique used to select the sample.**

**Code:**

```
sample = data.sample(n=30, random_state=42)
```

**8. Identify the numerical and categorical variables in the collected data.**

**9. Select one numerical variable for further analysis.**

**Code:**

```
data = pd.to_numeric(df.iloc[:, 8],
errors="coerce").dropna()
print("Valid observations:", len(data))
```

**Output:**

Valid observations: 33

**10. Calculate the sample size, mean, median, and standard deviation of the selected Variable.**

**Code:**

```
sample_size = len(sample)
sample_mean = sample.mean()
sample_median = sample.median()
sample_std = sample.std()
print("\nPART A - SAMPLE STATISTICS")
print("Sample Size:", sample_size)
print("Sample Mean:", sample_mean)
```

Aashu Yadav  
T050

```
print("Sample Median:", sample_median)
print("Sample Standard Deviation:", sample_std)
```

### Output:

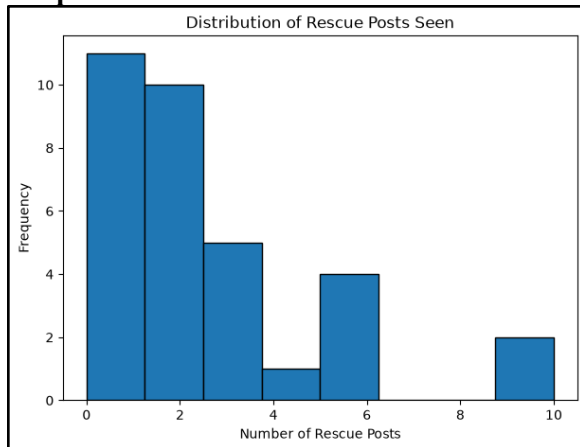
```
Sample Size: 30
Sample Mean: 2.4
Sample Median: 2.0
Sample Standard Deviation: 2.077797242107695
```

### 11. Create a suitable chart to display the collected data.

#### Code:

```
plt.figure(figsize=(7, 5))
plt.hist(data, bins=8, edgecolor="black")
plt.title("Distribution of Rescue Posts Seen")
plt.xlabel("Number of Rescue Posts")
plt.ylabel("Frequency")
plt.show()
```

### Output:



### 12. Write two observations from the survey data.

#### Observation 1:

Most respondents saw a limited number of bird or animal rescue posts on social media.

#### Observation 2:

The number of rescue posts seen varied among the respondents.

## Part B: Sampling Distribution and Central Limit Theorem

### 13. Select a numerical variable from the collected data.

Selected numerical variable: Q6 – Number of animal/bird rescue posts seen on social media in the past month.

### 14. Select a sample of 30 observations randomly from the data.

### 15. Calculate the mean of the selected sample.

### 16. Repeat the random sampling process 100 times.

### 17. Calculate and store the mean of each sample.

#### Code:

```
sample_size = 30
number_of_samples = 100
sample_means = []
# Repeat random sampling 100 times
for i in range(number_of_samples):
    random_sample = data.sample(
        n=sample_size,
        replace=True
    )
    mean = random_sample.mean()
    sample_means.append(mean)
```

### 18. Display the first 10 sample means.

#### Code:

```
print("\nFIRST 10 SAMPLE MEANS:")
for i in range(10):
    print("Sample", i + 1, "Mean =", sample_means[i])
```

### Output:

```
FIRST 10 SAMPLE MEANS:
Sample 1 Mean = 2.1
Sample 2 Mean = 2.6
Sample 3 Mean = 2.8666666666666667
Sample 4 Mean = 2.7
Sample 5 Mean = 3.0333333333333333
Sample 6 Mean = 2.5666666666666667
Sample 7 Mean = 2.3
Sample 8 Mean = 3.1666666666666665
Sample 9 Mean = 1.8333333333333333
Sample 10 Mean = 2.2333333333333334
```

### 19. Calculate the mean of all the sample means.

#### Code:

```
mean_of_sample_means = np.mean(sample_means)
print("\nMean of All Sample Means:",
      mean_of_sample_means)
# Original population mean
population_mean = data.mean()

print("\nPopulation Mean:",
      population_mean)
```

### Output:

```
Mean of All Sample Means: 2.6196666666666664
```

```
Population Mean: 2.606060606060606
```

### 20. Calculate the standard deviation of the sample means.

#### Code:

```
std_sampling_distribution = np.std(
    sample_means,
    ddof=1
)
print("Standard Deviation of Sample Means:",
      std_sampling_distribution)
```

### Output:

```
Standard Deviation of Sample Means: 0.40416171158752257
```

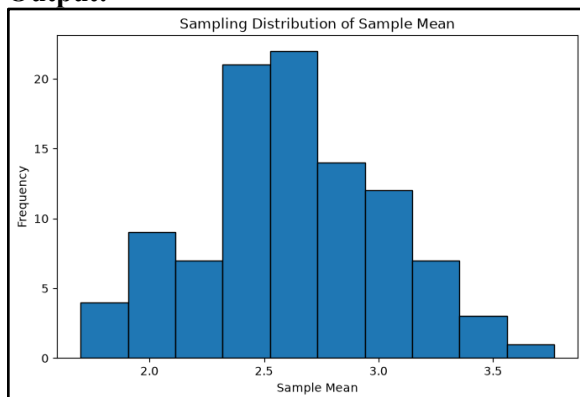
### Output:

### 21. Create a histogram of the sample means.

#### Code:

```
plt.figure(figsize=(8, 5))
plt.hist(
    sample_means,
    bins=10,
    edgecolor="black"
)
plt.title("Sampling Distribution of Sample Mean")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.show()
```

### Output:



### 22. Observe the shape of the sampling distribution.

The sampling distribution is approximately bell-shaped and concentrated around the central value.

### 23. Compare the original sample mean with the mean of the sampling distribution.

#### Code:

```
print("\nCOMPARISON")
print("Original Sample Mean:", sample_mean)
print("Mean of Sampling Distribution:",
      mean_of_sample_means)
print("\nPopulation Mean:", population_mean)
```

### Output:

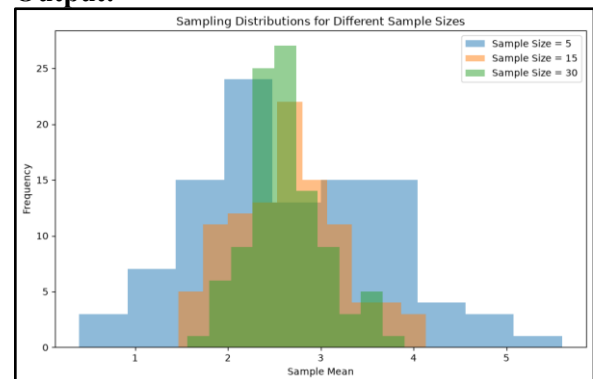
```
COMPARISON
Original Sample Mean: 2.4
Mean of Sampling Distribution: 2.6196666666666664
Population Mean: 2.606060606060606
```

### 24. Compare the sampling distributions obtained for different sample sizes.

#### Code:

```
sample_sizes = [5, 15, 30]
plt.figure(figsize=(10, 6))
for size in sample_sizes:
    means = []
    for i in range(100):
        random_sample = data.sample(n=size, replace=True)
        means.append(random_sample.mean())
    plt.hist(means, bins=10, alpha=0.5, label=f"Sample Size = {size}")
plt.title("Sampling Distributions for Different Sample Sizes")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

### Output:



### 25. Calculate the standard error of the sample mean.

#### Code:

```
standard_error = data.std(ddof=1) / np.sqrt(30)
print("\nPOINT 25 - STANDARD ERROR")
```

Aashu Yadav  
T050

```
print("Standard Error of Sample Mean:", standard_error)
```

**Output:**

```
Standard Error of Sample Mean: 0.43765329204794523
```

**26. Explain how the Central Limit Theorem is demonstrated through the experiment.**

The Central Limit Theorem was demonstrated using repeated random samples. The distribution of sample means became more concentrated as sample size increased.

**27. Use the sample mean to estimate the population mean.**

**Code:**

```
print("\nPOINT 27 - ESTIMATION OF POPULATION MEAN")
estimated_population_mean = sample_mean
print("Sample Mean:", sample_mean)
print("Estimated Population Mean:",
estimated_population_mean)
```

**Output:**

```
Sample Mean: 2.4
Estimated Population Mean: 2.4
```

**28. Write a conclusion based on the sampling distribution and Central Limit Theorem.**

Thus, sampling and sampling distribution were successfully performed using Python. The results demonstrated the Central Limit Theorem and showed that the sample mean can be used to estimate the population mean.